

CHARLES

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

A Computer Science undergraduate who believes technology should solve real human problems. Passionate about building scalable, maintainable, and reliable systems with a focus on clean architecture and long-term sustainability. Motivated to bring a result-driven mindset, strong collaboration, and practical problem-solving skills to deliver high-quality solutions.

EDUCATION

BINUS University

Jakarta, Indonesia

Bachelor of Computer Science, Specialization in Software Engineering – GPA: 3.80/4.00 (82 credits)

(2024 – 2028)

- Relevant Coursework: Algorithm and Programming (A-), Data Structure (A), Algorithm Design and Analysis (A-), Object Oriented Programming (A-), Artificial Intelligence (A), Database Technology (A), Software Architecture (A), Software Engineering (A), Code Reengineering (A)

Budi Agung Trilingual High School

Jakarta, Indonesia

High School Diploma, Mathematics and Natural Sciences

(2021 – 2024)

- Achieved 2nd highest score in writing a scientific paper with the title The Impact of Python Programming Language Scalability on Machine Learning Development.
- Achieved 2nd rank in the third year of high school.
- Achieved 4th rank in the second year of high school.

EXPERIENCES

S-Class Program BINUS @Kemanggisian

Jakarta, Indonesia

Sigma Lab Club – Product Development

(Feb 2026 – Present)

- Selected as 1 of 70 students from B2028 cohort for a product development track based on academic selection criteria.
- Completed weekly general and club training in product development fundamentals.
- Collaborated with cohort members in club-level training sessions to apply product development frameworks.

Keluarga Mahasiswa Buddhis Dhammavaddhana BINUS University

Jakarta, Indonesia

Secretary of Waisak Puja 2570 B.E./2026 X HUT KMBD XXXVII

(Jan 2026 – Aug 2026)

- Managed end-to-end administrative documentation for 35+ committee members across 5 divisions, including proposals, meeting minutes, MoAs, invitation letters, and venue/equipment permits, and accountability reports.
- Streamlined documentation workflow with AI-assisted automation tools.
- Coordinated external communications for venue booking, equipment logistics, and vendor arrangements.

Public Relations Coordinator of Waisak Puja 2569 B.E./2025 X HUT KMBD XXXVI

(Jan 2025 – Jul 2025)

- Led 6 committee members of Public Relations Division for Waisak Puja 2569 B.E./2025, Vesak Rhythm Fest, and HUT KMBD XXXVI, with 500+ participants.
- Coordinated internal and external communications, with 13 media partners and 30+ external organizations.
- Managed participant outreach through promotional content, event announcements, and reminder communications.

ISACA Student Group BINUS University

Jakarta, Indonesia

Project Manager of EXPO ISG 2026

(Jul 2026 – Sep 2026)

- Led 18 committee members across 2 divisions in organizing event booths.
- Managed end-to-end administrative and financial documentation, including proposals, meeting minutes, budget reports, and accountability reports.
- Increased registrants by approximately 83% year-over-year by expanding organizational outreach to BINUS @Bekasi.

BINUS University

Jakarta, Indonesia

Freshmen Partner B2029 BINUS @Kemanggisian

(Sep 2025 – Sep 2026)

- Mentored and monitored 8 freshmen throughout the one-year Excellence Program (EP).
- Conducted 13 weekly sessions to introduce and discuss BINUS Graduate Attributes (BGA) with freshmen.
- Guided freshmen in adapting to the academic and social environment at university life.

Team Promotion and Events

(Jul 2025 – Dec 2025)

- Promoted university programs to prospective students across multiple school visits.

- Supported event setup, logistical preparation, and on-site coordination for promotional activities.
- Assisted lecturers during school visits, seminars, and university promotional activities.

Freshmen Leader B2029 BINUS @Kemanggisan

(Jul 2025 – Sep 2025)

- Facilitated orientation sessions and group discussions for 70+ freshmen.
- Guided and monitored 8 freshmen throughout the New Student Experience and Transition (NEXT) program.
- Introduced the Computer Science program curriculum, academic resources, and practical strategies for university life.

PROJECTS

HapHap – Surplus Food Marketplace

(2026)

- Designed user flows, wireframes, and high-fidelity mobile-first interfaces for a surplus food marketplace.
- Developed a full-stack mobile application MVP using Flutter, NestJS, PostgreSQL, Supabase, and Midtrans.
- Architected a scalable, maintainable, and reliable layered client-server architecture for semi-production-ready deployment.

Klean – Laundry Marketplace

(2026)

- Applied Scrum/Agile and SDLC with an AI-driven workflow across product planning, design, development, and iteration.
- Designed user flows, wireframes, and high-fidelity mobile-first interfaces for a cross-platform laundry marketplace PWA.
- Developed a full-stack web application MVP using React, Vite, Express.js, PostgreSQL, Supabase, and Midtrans.

Bersih.In – AI Waste Classifier

(2025)

- Trained an EfficientNet-B0 image classification model on 45,000+ waste images, with approximately 98% validation accuracy.
- Designed user flows, wireframes, and high-fidelity interfaces for a single-page AI waste classification web application.
- Developed a full-stack web application MVP using React, Vite, Express.js, and Flask for model inference.

Katasaka – AI-Integrated Folklore and Local Language Platform

(2025)

- Designed user flows, wireframes, and high-fidelity mobile-first interfaces for an AI-integrated folklore and local language platform.
- Conducted market validation through a survey of 50+ respondents, with approximately 75% confirming the identified problem.
- Selected as Top 50 Semifinalist at DigiHack 2025 national hackathon competition organized by Telkom University.

RESEARCH

Comparative Study of Transformer Models for Indonesian Multi-Label Hate Speech and Abusive Language Detection **(2026)**

The 13th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI) 2026

- Designed and fine-tuned a controlled benchmark of four transformer models (IndoBERT, IndoBERTweet, mBERT, and XLM-RoBERTa) across 8 experimental runs on 13,169 Indonesian tweets with 12 binary labels.
- Identified IndoBERT as the best-performing model with a macro F1-score of 0.7036, with monolingual models outperforming multilingual models and preprocessing effects varying by architecture.
- Conceptualized the study, implemented the experiments, drafted the manuscript, and produced research visualizations.

A Dual Explainability Framework for Cardiovascular Disease Prediction Using SHAP and LIME with Tree-Based Machine Learning Models **(2026)**

The 13th International Conference on ICT for Smart Society (ICISS) 2026

- Designed and optimized three tree-based ensemble models (Random Forest, XGBoost, and LightGBM) on 918 clinical records.
- Identified Random Forest as the best-performing model with a cross-validation recall of 0.9112 and found consistent global and patient-level explanations using SHAP and LIME.
- Designed the research methodology, implemented the models, performed data preprocessing and optimization, and conducted SHAP and LIME analysis.

OTHERS

Languages: English (Professional), Chinese (Elementary), Indonesian (Native)

Programming Languages: TypeScript, JavaScript, Python, Java, Kotlin, XML, Dart, C

Frameworks & Libraries: React, Tailwind, Axios, Express.js, NestJS, Prisma, Flutter

Databases & Infrastructure: MySQL, PostgreSQL, Supabase, Vercel, Railway

Tools & Methodology: Agile/Scrum, Visual Studio Code, Git, GitHub, Postman, Figma